

Lesson 3-1

The Overall Structure of Web Applications

Learning Objectives

- Explain what a web application is and describe the roles of its three tiers: frontend, backend, and database.
- Trace how the three tiers cooperate to turn a user's action into a result, using familiar services as examples.

Every time you search on YouTube, buy on Amazon, or open Gmail, you are using a web application — a program you use through your browser, with nothing to install. Behind the simple screen, three tiers work together: the page you interact with, the hidden server that processes the request, and the database that holds all the data. Understanding these three tiers explains how almost every online service is built.

? Guiding question: How is a web application structured, and how do its three tiers work together to respond to what a user does?

Point!

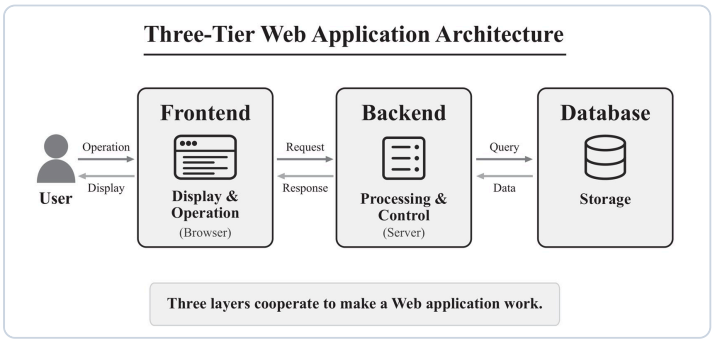
1. What is a Web Application?

(1) **(Web application)**…an application used through a web browser. It can be used as long as you have a browser, without needing to install dedicated software.
e.g., YouTube, Amazon, Gmail, Google Maps

2. The Three Tiers of a Web Application

(1) A web application is broadly composed of the following three tiers.

Tier	Role	Examples
Frontend	The screen component that users see directly. It runs on the browser and accepts user input.	Search screen, product listing pages, input forms, etc.
Backend	The server-side processing that users do not see. It processes data and makes decisions.	Searching for products that match search criteria, login authentication, etc.
Database	A system that organizes, stores, and manages data. It retrieves data when needed and records new data.	Product information, user information, order history, etc.



Three-Tier Web Application Architecture

3. The Flow of How the Three Tiers Cooperate

(1) When a user operates a web application, the three tiers cooperate to perform the processing to make it work.

Key Fact:
A web application is built from three tiers — frontend, backend, and database — that cooperate to turn a user's action into a result.

Key Concepts:

- web application
- frontend
- backend
- database
- three-tier architecture

KEY TERMS
web application — an application used through a web browser, with no dedicated software to install.



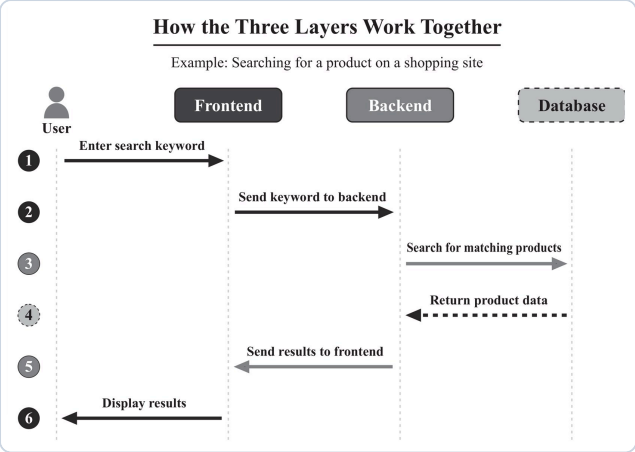
The data tier stores and serves the records.



Three tiers: presentation, logic, and data.

e.g., Searching for a product on an online shopping site

① The frontend receives the search keywords entered by the user.
② The frontend sends the search keywords to the backend.
③ The backend searches the database for products that match the criteria.
④ The database returns the data of matching products to the backend.
⑤ The backend organizes the search results and returns them to the frontend.
⑥ The frontend displays the search results on the user's screen.



How the Three Layers Work Together

4. Familiar Web Application Configurations

(1) All familiar web applications also operate on this three-tier architecture.

Service	Frontend (display and interaction)	Backend (processing and control)	Database (storage and management)
Video sharing site	Display of video playback and search screens; reception of user operations	Selection of recommended videos; aggregation of view counts	Storage of video data, user information, and viewing history
SNS	Display of timelines and post screens; sending of input content	Determining the audience for posts; control of push notifications	Storage of post data, follow relationships, and messages
E-commerce site	Display of product listings and cart screens; reception of order operations	Stock checking; execution of payment systems	Storage of product information, order history, and customer information

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A The frontend is the screen component that users see directly.

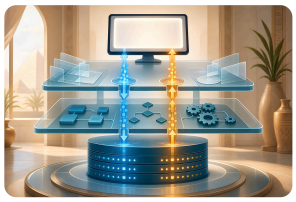
B The backend is the part that runs on the browser.

C A database is a system that organizes, stores, and manages data.

D A web application operates with the frontend alone.

PAUSE & THINK

When you type in a search box and see results, which tier collected your input, and which tier found the answer?



A request flows down the tiers and a reply returns.

KEY TERMS

frontend — the screen users see and interact with.
backend — the hidden server-side processing.
database — where data is stored and managed.

PAUSE & THINK

In the table, every service uses the same three tiers — what changes from one service to the next?

(2) Match each description (a–c) with the most appropriate term from the choices below (A–C).

a The part responsible for server-side processing that users do not see

b The screen component that users interact with directly

c A system that organizes, stores, and manages data

[Choices] **A** Frontend **B** Backend **C** Database

✓ Solution

(1)

A The frontend is the screen component that users see directly. Therefore, ○.

B The backend is the part that runs on the server, not on the browser. Therefore, ×.

C Correct definition of a database. Therefore, ○.

D A web application operates with three tiers: frontend, backend, and database. Therefore, ×.

(2)

a: Server-side processing is the backend. **B**

b: The screen that users interact with is the frontend. **A**

c: Storing and managing data is the database. **C**

Try

1 Answer the following questions.

1. What is the term for an application used through a web browser?
2. What is the term for the screen component that users see directly?
3. What is the term for the part responsible for server-side processing that users do not see?
4. What is the term for a system that organizes, stores, and manages data?

2 Answer the following questions.

(1) When searching for a product on an online shopping site, which of the following is responsible for searching through data for products that match the search keywords: **A** Frontend, **B** Backend, **C** Database?

EXAM-STYLE QUESTION

Explain how the three tiers of a web application cooperate when a user posts a photo on a social media app. **[6] Refer to:** the role of the frontend, backend, and database.

(2) For each of a–f, indicate which tier of an e-commerce site it belongs to: **A** Frontend, **B** Backend, **C** Database.

a Product listing page

b Stock checking processing

c Storage of customer information

d Cart screen

e Payment processing

f Storage of order history

⚙ Think as an Engineer — Investigate & Decide

A clinic wants a web app where patients book appointments, check which time slots are free, and see their past visits. You plan its three tiers.

☐ **Put everything in one program**

☐ **Separate the frontend, backend, and database**

1 · Assign the work. For three features — the booking form, checking which time slots are free, and storing past visits — say which tier (frontend, backend, or database) should handle each, and why.

2 · Diagnose a fault. The booking page opens normally, but no free slots appear on screen. Which tier is clearly working, and which one has probably failed? Explain your reasoning.

3 · Decide. Recommend whether the clinic should separate its app into three tiers, and give two reasons. **In pairs,** compare your assignments and agree together on where checking free time slots belongs.

Give evidence for your answer.

Stuck? The frontend shows and collects; the backend decides and processes; the database stores and retrieves.

🌍 In a New Context

A weather app lets users type a city and see the forecast. Using the three-tier model, name which tier does each job — taking the city name, looking up the forecast data, and storing past forecasts — and identify one job the frontend cannot do on its own.

KEY TERMS

three-tier architecture — frontend + backend + database cooperating.

❓ Lesson Question — Answered

The lesson asked how a web application is structured and how its tiers cooperate. A web application runs in a browser and is built from three tiers. The frontend is the screen the user sees and interacts with; the backend is the hidden server-side processing that makes decisions; and the database organizes, stores, and manages the data. When a user acts — for example, searching for a product — the frontend collects the input and passes it to the backend, the backend queries the database, the database returns the data, and the backend sends the organized result back to the frontend to display. Overall, a web application works not through one part but through three tiers cooperating, each with a clear role.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

A web application is broadly composed of three tiers. The screen component that users see directly is called the (a). The part responsible for server-side processing that users do not see is called the (b). A system that organizes, stores, and manages data is called the (c). In response to user actions, these three tiers cooperate to perform processing.

1. Fill in the blanks (a)–(c).

3 Answer the following questions.

(1) For each of a–f, indicate which tier of an SNS it belongs to: **A** Frontend, **B** Backend, **C** Database.

a Timeline screen
b Processing that determines the audience for a post
c Storage of post data
d Post screen
e Notification processing
f Storage of follow relationships

⭐ REFLECT & CHALLENGE

Reflect: Which tier do you think fails most noticeably to a user when something goes wrong, and why? **Challenge:** Name one online service you use and describe what its frontend, backend, and database each do.

⭐ Key takeaway

Remember:
Frontend shows and collects, backend decides and processes, database stores and retrieves — three tiers cooperating make a web application work.

(2) From the following options (A–D), choose the one that most appropriately describes the three-tier architecture of a web application.

A The frontend has the role of storing and managing data.

B The backend is the part that users interact with directly on the browser.

C In response to user actions, the frontend, backend, and database cooperate to perform processing.

D The database has the role of displaying screens to users.